

PNG 515.1:

Formation Damage Assessment and Control

OBJECTIVES

At the end of this lecture, students should be able to:

- ❖ Diagnose production decline using engineering tools
- ❖ Differentiate natural reservoir depletion from formation damage
- ❖ Explain well performance concepts (IPR, PI, Skin, Nodal Analysis)
- ❖ Identify types and causes of formation damage
- ❖ Recognize symptoms of formation damage
- ❖ Design formation damage control strategies
- ❖ Explain stimulation washes and matrix acidization in detail

PART 1: INTRODUCTION TO FORMATION DAMAGE

What is Formation Damage?

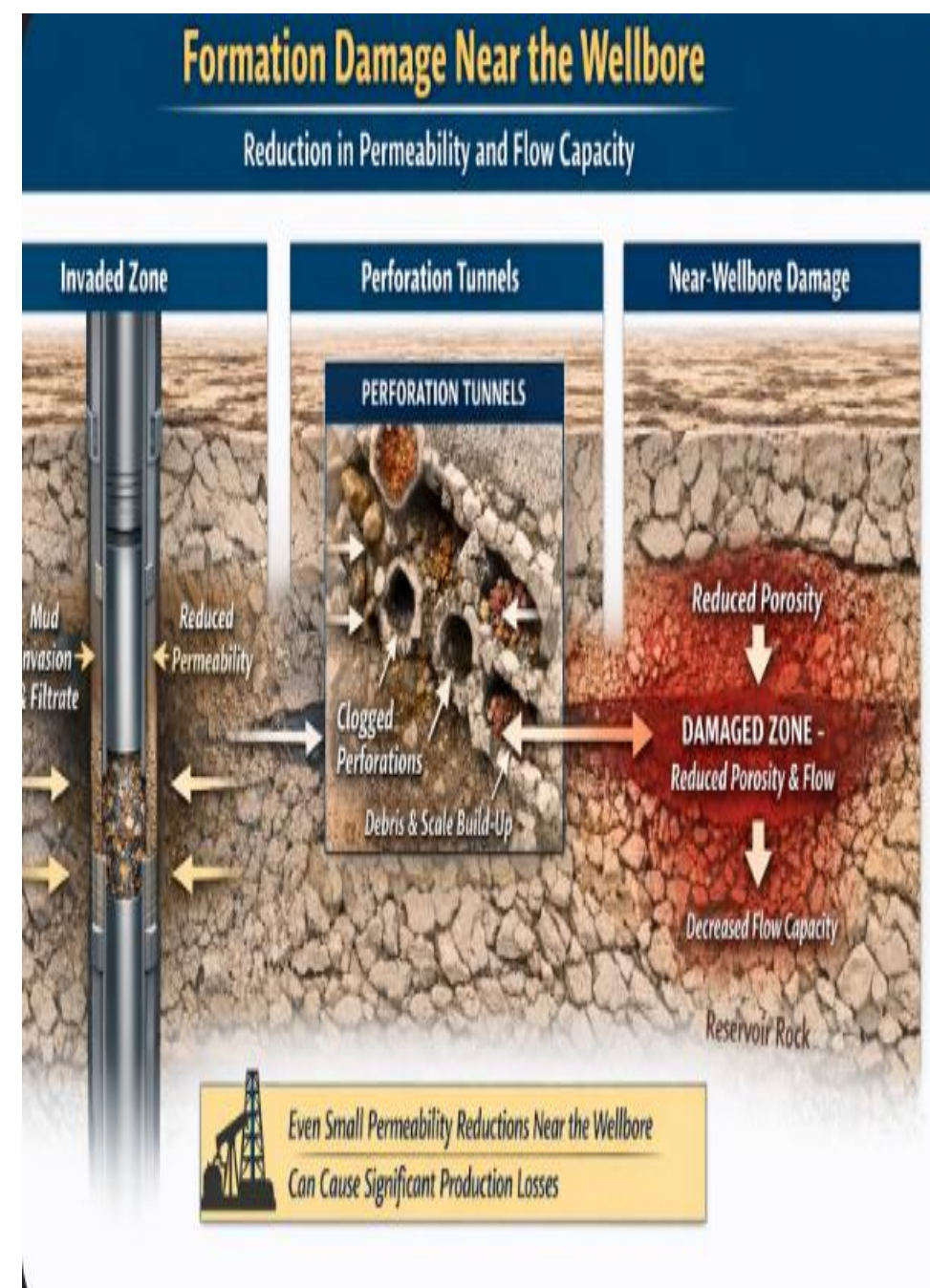
Formation damage is any reduction in permeability or flow capacity in the near-wellbore region caused by mechanical, chemical, biological, or operational processes.

It occurs within:

- The invaded zone
- The perforation tunnels
- The immediate reservoir rock around the wellbore

Key Engineering Concept:

Even small permeability reductions near the wellbore can cause significant production losses because flow convergence is highest near the well.



TYPES AND CAUSES OF FORMATION DAMAGE



Mechanical Damage

(a) Fines Migration

Triggered by:

- High drawdown
- Water breakthrough
- Acid treatments

Fines plug pore throats.

(b) Sand Production

Occurs in unconsolidated formations.

(c) Compaction Damage

Reservoir stress changes reduce permeability.

Chemical Damage

(a) Scale Formation

Common scales:

- CaCO_3
- BaSO_4
- CaSO_4

Occurs due to:

- Mixing incompatible waters
- Pressure/temperature changes

(b) Clay Swelling

Smectite clays swell in fresh water.

Results:

- Pore throat blockage
- Reduced permeability

(c) Emulsion Blockage

Oil-water emulsions block pores.

(d) Polymer residues

Biological Damage

- Bacterial growth
- Biofilm formation

Common in water injection wells.

Completion-Induced Damage

- Cement filtrate
- Perforation debris
- Improper cleanup

Drilling-Induced Damage

- Solids invasion
 - Filter cake buildup
 - Mud filtrate invasion
- Water-based muds can cause clay swelling.

SYMPTOMS OF FORMATION DAMAGE

- ❖ Sudden production drop
- ❖ Increasing drawdown for same rate
- ❖ Reduced productivity index
- ❖ High skin factor
- ❖ Distorted pressure buildup curve
- ❖ Increase in water cut without reservoir explanation
- ❖ Poor response to artificial lift

WELL DIAGNOSTICS & DECLINE DIAGNOSIS

- Well diagnostics refers to the systematic evaluation of a well's performance to determine why production is declining and to identify the underlying cause.
- Decline in production is inevitable in petroleum reservoirs, but the key engineering challenge is distinguishing between natural reservoir depletion and correctable well problems.

Decline diagnosis integrates:

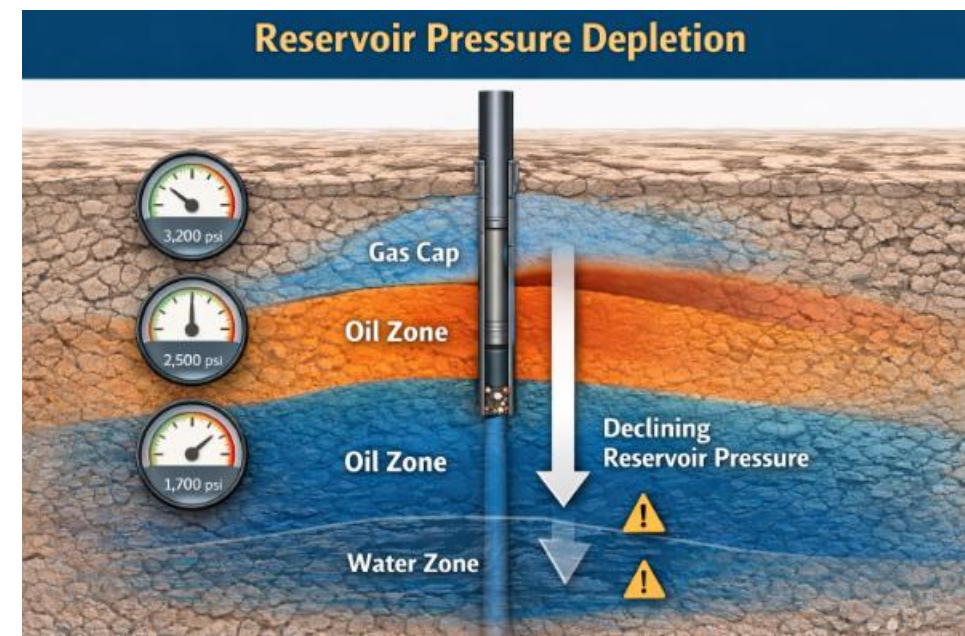
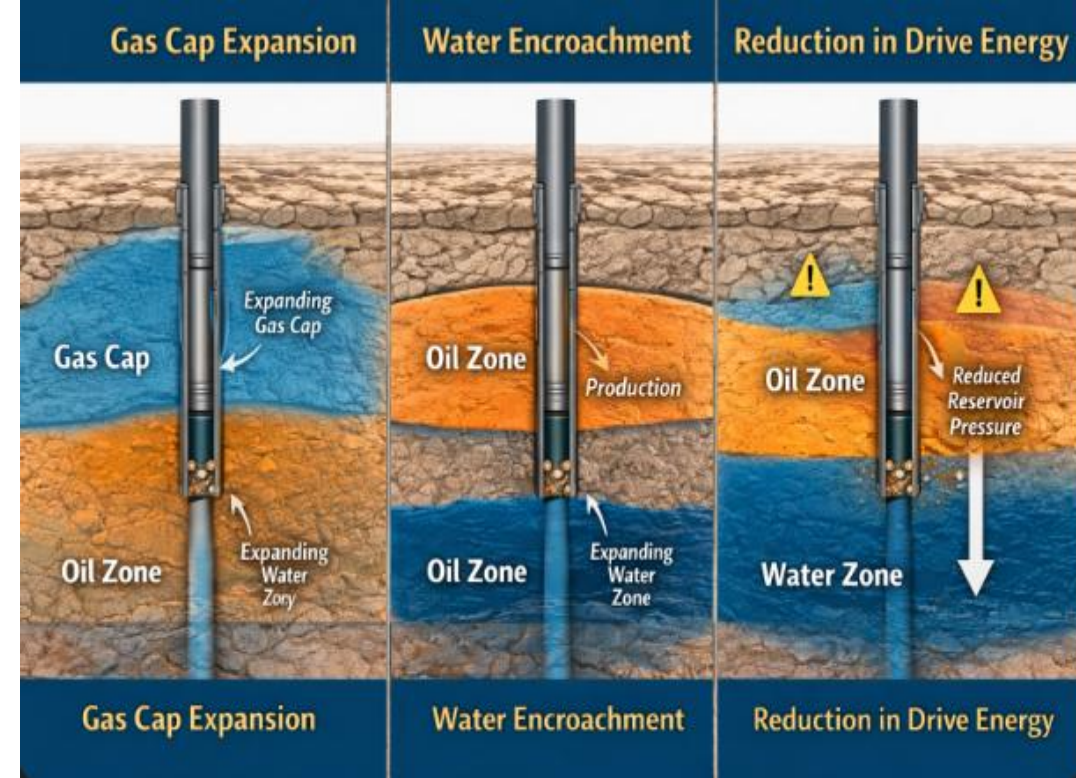
- Reservoir engineering
- Production engineering
- Petrophysics
- Well testing
- Completion diagnostics

Natural Production Decline

Production decline occurs due to:

- ❖ Reservoir pressure depletion
- ❖ Gas cap expansion
- ❖ Water encroachment
- ❖ Reduction in drive energy

These are expected reservoir behaviors and are analyzed using decline curve analysis.



Decline Curve Analysis (DCA)

Decline curve analysis was introduced by J.J. Arps in 1945.

Arps identified three main decline types:

1. Exponential Decline

Occurs when decline rate is constant.

$$q = q_i e^{-Dt}$$

q = Production rate at time t (STB/day or MSCF/day)

e = Mathematical constant (Euler's number ≈ 2.718)

D (Decline Rate) = Constant fractional decline rate per unit time (1/year or 1/month)

q_i (Initial Production Rate) = Production rate at time t = 0

t = Time since start of decline (years or months)

2. Hyperbolic Decline

Decline rate decreases with time.

$$q = \frac{q_i}{(1 + bDt)^{1/b}}$$

3. Harmonic Decline

Special case of hyperbolic (b = 1).

Comparison Table

Feature	Exponential	Harmonic	Hyperbolic
Equation	$q = q_i e^{-Dt}$	$q = \frac{q_i}{1 + Dt}$	$q = \frac{q_i}{(1 + bDt)^{1/b}}$
Decline Rate	Constant	Decreasing	Decreasing
Decline Exponent (b)	0	1	$0 < b < 1$
Early-Time Drop	Moderate	High	High
Late-Time Tail	Short	Long	Long
Forecast Conservatism	Most conservative	Most optimistic	Moderate
Plot Type	Straight line (semi-log)	Curved	Curved

If:

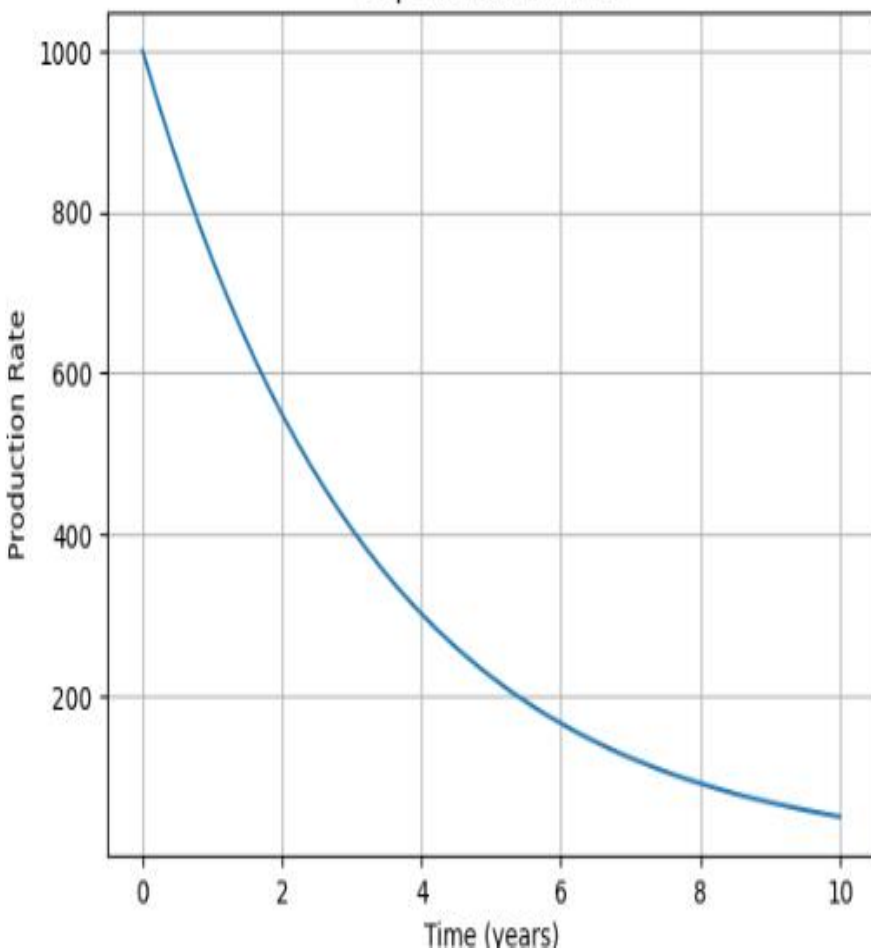
• $b=0 \rightarrow$ Exponential decline

• $b=1 \rightarrow$ Harmonic decline

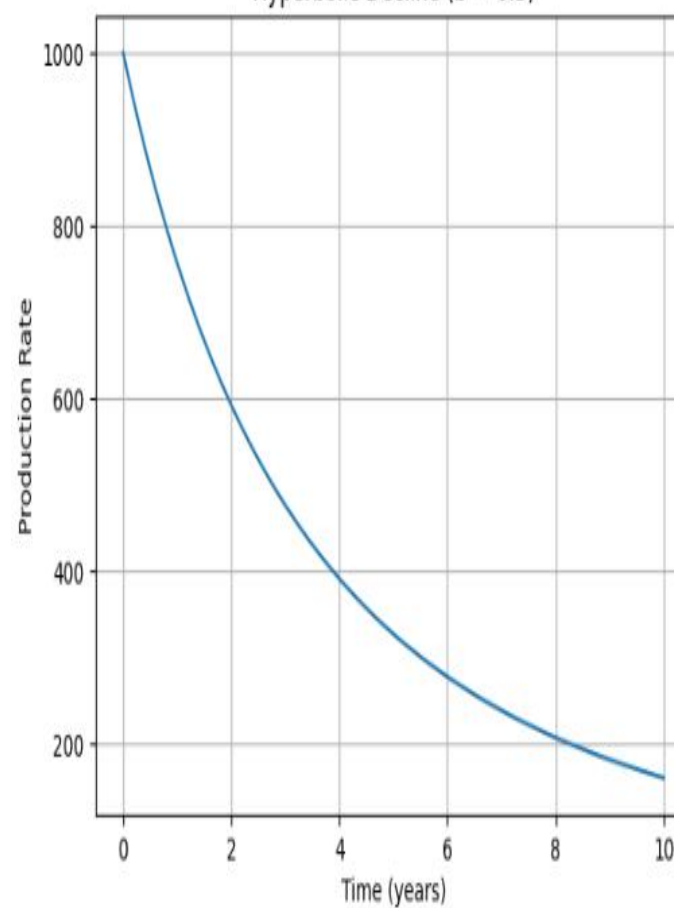
• $0 < b < 1 \rightarrow$ Hyperbolic decline

Thus:
Exponential and Harmonic are special cases of Hyperbolic decline.

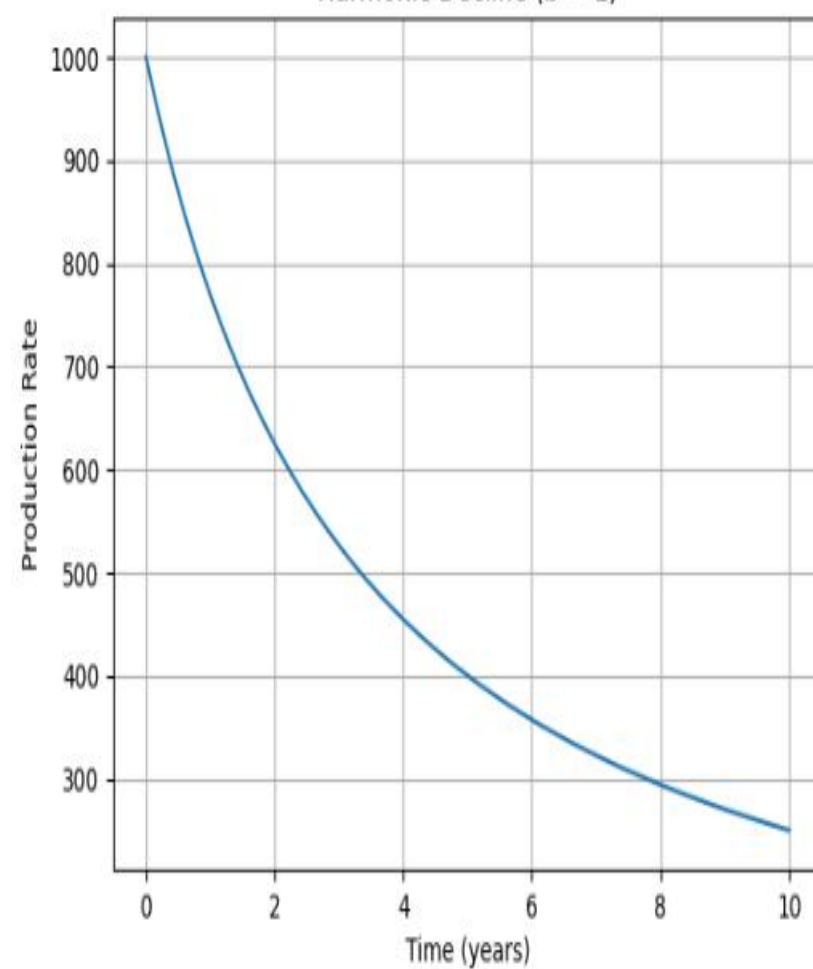
Exponential Decline



Hyperbolic Decline ($b = 0.5$)



Harmonic Decline ($b = 1$)



Engineering Implications in Well Diagnostics

If decline follows exponential:

Likely normal depletion and boundary-dominated flow.

If decline follows hyperbolic:

Likely transient flow or fracture-dominated behavior.

If decline follows harmonic:

Indicates strong transient effects or complex reservoir behavior.

Engineering Insight

- ❖ Exponential is most conservative for reserves estimation.
- ❖ Hyperbolic and harmonic can overestimate reserves if applied incorrectly.
- ❖ Proper model selection is critical in:
 - Well diagnostics
 - Reserve estimation
 - Economic forecasting
 - Production planning

Final Summary

- Exponential: constant percentage decline
- Harmonic: decline slows significantly over time
- Hyperbolic: intermediate case with decreasing decline rate

All three models originate from the empirical work of J.J. Arps and remain foundational tools in petroleum production engineering.

Interpretation for Diagnostics

If production decline follows Arps' predicted trend → likely natural depletion.

If production drops sharply or deviates from expected trend → possible:

- Formation damage
- Mechanical failure
- Completion issues
- Scale or wax deposition

Well Diagnostic Tools

1. Pressure Transient Testing

- Drawdown test

- Build-up test

- Interference test

Skin factor (S) is key:

- $S \approx 0 \rightarrow$ no damage

- $S > 0 \rightarrow$ formation damage

- $S < 0 \rightarrow$ stimulation successful

2. Production Logging Tool (PLT)

Measures:

- Flow profile

- Water entry

- Gas breakthrough

Used to detect:

- Channeling

- Water coning

- Crossflow

Well Logging

Examples:

- Resistivity logs
- Neutron-density logs
- Sonic logs

Used to identify:

- Water saturation changes
- Sand production
- Compaction

Surface Diagnostic Indicators

- Increase in water cut
- Rising GOR
- Reduced flowing pressure
- Increased tubing head pressure

WELL PERFORMANCE FUNDAMENTALS

Inflow Performance Relationship (IPR): IPR describes the relationship between reservoir pressure and production rate.

For oil wells (Vogel equation): $\frac{q}{q_{max}} = 1 - 0.2 \left(\frac{P_{wf}}{P_r} \right) - 0.8 \left(\frac{P_{wf}}{P_r} \right)^2$

For gas wells (deliverability equation): $q = C (P_r^2 - P_{wf}^2)^n$

Where:

- q = production rate
- Pr = reservoir pressure
- Pwf = flowing bottom-hole pressure

Skin Effect on Productivity: $PI = \frac{0.00708kh}{\mu B [\ln(r_e/r_w) + S]}$

Higher skin → lower PI

Example:

If S increases from 0 to 5, production can drop by 40–60%.

Outflow Performance (Tubing Performance)

Represents pressure losses in tubing due to:

- Friction
- Hydrostatic head
- Acceleration

Nodal Analysis

Combines IPR and tubing performance to determine:

- Operating rate
- Bottom-hole pressure
- Artificial lift requirement

If nodal intersection shifts downward → possible formation damage.

Causes of Increased Skin

- Mud invasion
- Clay swelling
- Scale deposition
- Emulsion blockage
- Fines migration

FORMATION DAMAGE CONTROL

Control strategy must begin during:



Drilling Phase

- Use compatible mud system
- Optimize solids control
- Maintain low fluid loss
- Avoid overbalance damage

Completion Phase

- Clean perforations
- Use filtered completion fluids
- Minimize perforation debris
- Control drawdown

Production Phase

- Use scale inhibitors
- Use corrosion inhibitors
- Control water injection chemistry
- Manage drawdown

REMEDIAL MEASURES

1. Stimulation washes
2. Matrix acidization
3. Hydraulic fracturing
4. Solvent treatment
5. Re-perforation

What is Well Stimulation?

❖ It is a well intervention performed on an oil or gas well to increase production by improving the flow of hydrocarbons from the reservoir into the well bore

such as:

1. Acidizing
2. fracturing

➤ They can be conducted on old wells and new wells and they can be also designed for remedial purposes.

In more recent times, due to the temporary nature of well stimulation, specialized drilling ships known as "well stimulation vessels" have been used for deep sea well stimulation.

What Are Stimulation Washers?

- ❖ They are specialized tools or systems used in oil and gas well operations to clean or stimulate the formation near the wellbore.
- ❖ Stimulation washers play a crucial role in formation damage control by helping to restore or enhance the permeability of the formation, ensuring better well productivity.
- ❖ They aid in cleaning the wellbore, preventing the invasion of debris, controlling clay damage, and optimizing the effectiveness of stimulation techniques such as acidizing and fracturing.
- ❖ However, their use must be carefully monitored, and the right fluids and methods should be selected to avoid worsening damage.
- ❖ Understanding the correct application of stimulation washers is essential for successful well stimulation and formation damage prevention, which can ultimately lead to more efficient and cost-effective hydrocarbon recovery

They are typically used during:

- i) Well stimulation: Techniques such as acidizing, hydraulic fracturing, or sand control operations.
 - ii) Well clean-up: Removing drilling muds, debris, and other contaminants that may block the flow of fluids from the reservoir.
- ❖ Stimulation washers work by injecting various fluids (e.g., water, acid, or solvents) into the wellbore and the surrounding formation to flush out debris, disperse harmful particles, and improve the permeability of the reservoir.

How Stimulation Washers Work

❖ Stimulation washers utilize various mechanisms to prevent or mitigate formation damage during well operations:

1. Mechanical Cleaning:

Ø Some stimulation washers function by physically removing debris and mud filtrates that may have entered the formation during drilling or previous operations.

Ø This is done using high-velocity fluid injection that helps dislodge particles that have adhered to the formation surface or become trapped in the pore spaces.

2. Chemical Dispersion: Other stimulation washers use chemicals such as solvents or surfactants that break up mud cakes, emulsions, or other contaminants. These chemicals dissolve or disperse particles that may have accumulated in the formation, helping to restore permeability

3. Clay Stabilization: When clay swelling is a concern, stimulation washers can be used to inject fluids that stabilize the clays and prevent them from swelling and migrating into the formation. This can involve the use of specific inhibitors, such as salt solutions, that reduce the potential for swelling.

4. Acidizing: In formations with calcite, dolomite, or other carbonate minerals, stimulation washers may inject acidizing agents (such as hydrochloric acid) to dissolve scale or damage caused by the interaction of the reservoir rock with other fluids.

5. Flowback of Damage-Inducing Fluids: In some cases, stimulation washers help with the controlled flowback of damage-inducing fluids (e.g., completion brine, drilling mud) from the wellbore into the surface, which can prevent these fluids from causing long-term damage to the formation.

Role of Stimulation Washers in Formation Damage Control

1. **Preventing Mud Invasion:** During drilling and completion, mud invasion into the formation can block the natural pores of the rock, leading to significant permeability reduction. Stimulation washers help to clear these muds and improve the permeability of the near-wellbore region.
2. **Improving Acid Stimulation Efficiency:** Acid treatments, particularly in carbonate formations, are often used to improve well productivity. Stimulation washers ensure that the acid is evenly distributed and can reach deeper into the formation, dissolving scales, and removing debris.
3. **Clay Damage Control:** Clay swelling and fines migration are common causes of formation damage. Stimulation washers inject fluids that stabilize the clays and prevent their migration, preserving the permeability of the formation.
4. **Cleaning Permeability Zones:** A major aspect of formation damage control is cleaning and restoring the formation's permeability. Stimulation washers help by flushing out debris, drilling fluids, and other contaminants from the formation's pore spaces, ensuring that the natural flow of fluids can be re-established.

5. Enhancing Hydraulic Fracturing: In hydraulic fracturing treatments, stimulation washers play a role in preventing damage caused by the proppants and fluids injected into the formation. The use of washers during fracturing ensures that particles do not clog or damage the formation and that fracturing fluid can flow easily

Applications of Stimulation Washers

1. **Acidizing Treatments:** When acidizing is used to enhance well productivity, stimulation washers can be employed to clear any obstructions in the near-wellbore zone and enhance the acid's penetration and effectiveness.
2. **Fracturing Operations:** In hydraulic fracturing, stimulation washers help by ensuring that the fracturing fluid and proppant can flow effectively into the formation without creating any new damage or blockages.
3. **Sand Control:** For reservoirs prone to sand production, stimulation washers can help by clearing any obstructing materials and improving the formation's ability to hold the injected fluids and proppants.
4. **Wellbore Clean-up:** After drilling, completion, or workover activities, stimulation washers are often used to clean the wellbore and surrounding formation, removing debris that could interfere with fluid flow and reducing the chances of further damage during production.

Acidizing

- The pumping of acid into the wellbore to remove near-well formation damage and other damaging substances.
 - This procedure commonly enhances production by increasing the effective well radius.
- Ø The two basic types of acidizing are characterized through injection rates and pressures:
1. Injection rates below fracture pressure are termed **Matrix acidizing**.
 2. Injection rates above fracture pressure are termed **Fracture acidizing**

■ MATRIX ACIDIZATION

Matrix acidization is a stimulation technique where acid is injected below fracture pressure to dissolve formation damage and increase permeability.

Objectives

- ❖ Remove drilling mud damage
- ❖ Dissolve carbonate scale
- ❖ Improve permeability
- ❖ Reduce skin factor

Acid Types

1. Hydrochloric Acid (HCl)

Used in carbonate formations.

Reaction with CaCO_3 : $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$

Mud Acid (HCl + HF)

Used in sandstone formations.

HF dissolves:

- Silica

- Clays

Organic Acids

- Acetic acid

- Formic acid

Used in high-temperature wells.

Acidizing Design Considerations

- Mineralogy of formation

- Temperature

- Formation permeability

- Acid concentration

- Additives (corrosion inhibitors, iron control agents)

Acidizing Procedure

- Pre-flush (remove incompatible ions)
- Main acid stage
- Overflush (push acid into formation)

Risks

- Acid precipitation
- Fines migration
- Corrosion
- Secondary damage

COMPARISON: MATRIX ACIDIZING VS FRACTURING

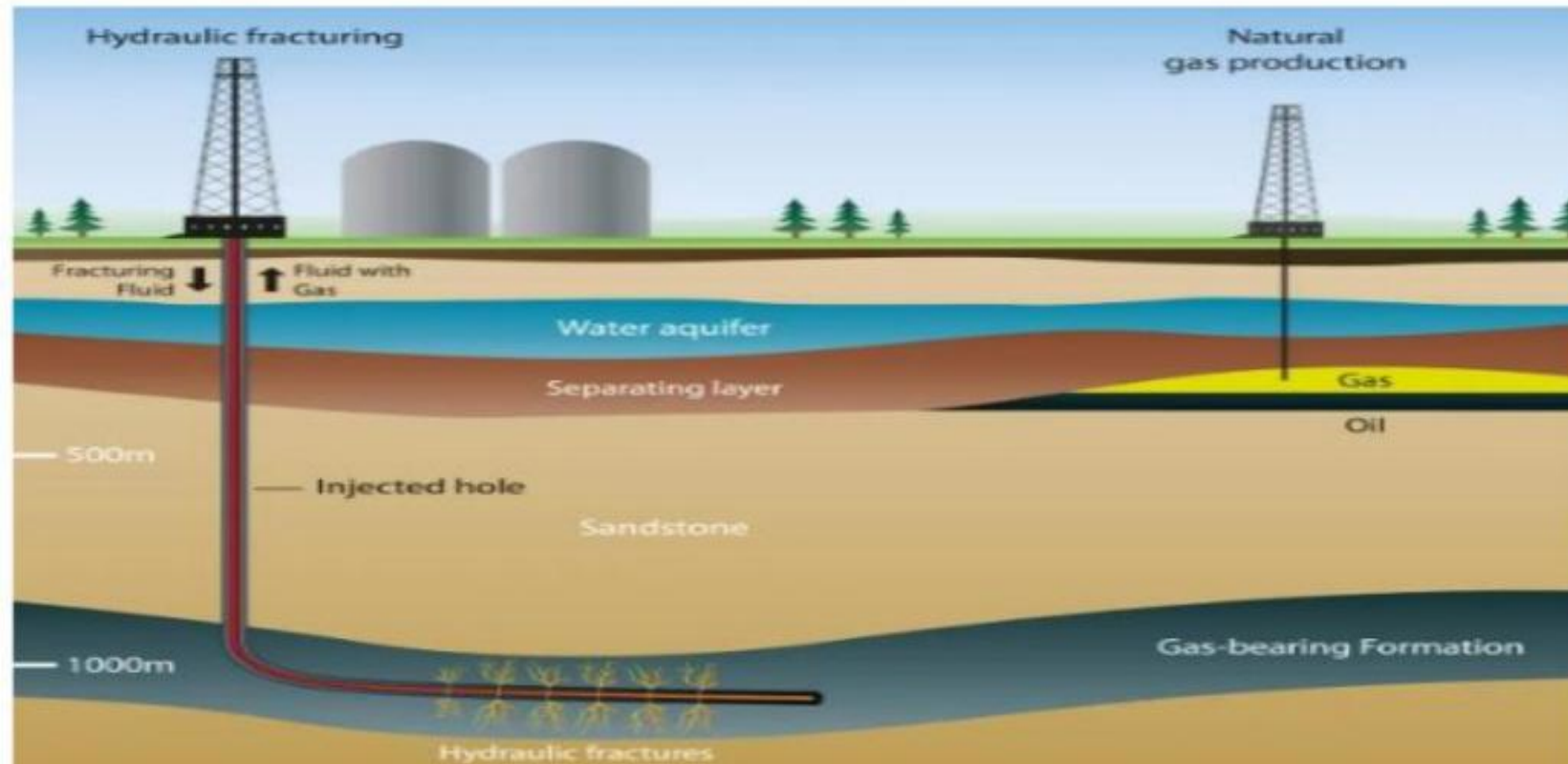
Parameter	Matrix Acidizing	Hydraulic Fracturing
Injection Pressure	Below fracture pressure	Above fracture pressure
Objective	Remove damage	Create new flow paths
Suitable for	Mild damage	Tight formations
Cost	Lower	Higher

Fracture acidizing

- Fracture acidizing is an alternative to hydraulic fracturing and propping in carbonate reservoirs. In fracture acidizing, the reservoir is hydraulically fractured and then the fracture faces are etched with acid to provide linear flow channels to wellbore.
- As such, the application of acid fracturing is confined to carbonate reservoirs and should never be used to stimulate sandstone, shale, or coal-seam reservoirs.
- Long etched fractures are difficult to obtain, because of high leak off and rapid acid reaction with the formation
- Fracking is the most widely used well stimulation technique particularly for low-permeability rock formations like shale.
- It involves injecting a high-pressure fluid (fracturing fluid) mixed with proppant (sand-like particles) into the wellbore.

- The fracturing fluid, typically a water-based solution with additives, creates fractures in the rock formation by overcoming the rock's natural pressure.
- Proppant, usually ceramic or sand, keeps these fractures open after the pressure is released, creating a permanent pathway for oil and gas to flow more easily towards the wellbore.

Hydraulic Fracturing (Fracking)



Other Well Stimulation Methods

Beyond fracking and acidizing, several other stimulation techniques exist for specialized applications:

- **Nitrogen Lift:** This method utilizes pressurized nitrogen gas injected into the wellbore to remove debris, lighten wellbore fluids, and improve well productivity. Nitrogen gas, being much lighter than formation fluids, helps to lift fluids and enhance well flow, particularly in wells with low reservoir pressure or high fluid viscosity.
- **Jetting:** High-pressure jets of water or abrasive fluids can be used to clean perforations or remove near-wellbore formation damage. This technique is particularly effective for removing drilling mud residue or formation debris that can clog perforations and restrict flow

Choosing the Right Well Stimulation Method

1. Reservoir Geology:

Rock Type: Sandstone formations often benefit from acidizing to dissolve mineral deposits like calcium carbonate. Shale formations, on the other hand, might require hydraulic fracturing to create and maintain flow channels due to their naturally low permeability.

Permeability: Formations with extremely low permeability, like tight sandstones or shales, might necessitate a combination of techniques. For instance, matrix acidizing could be used alongside hydraulic fracturing to create a network of conductive pathways within the rock matrix.

Formation Damage: Identifying the type and location of formation damage is essential. Acidizing is well-suited for removing carbonate deposits, while mechanical techniques like jetting might be better suited for clearing debris near the wellbore.

2. Wellbore Completion

Casing and Cement Integrity: The strength and integrity of the wellbore casing are crucial for withstanding the high pressures associated with hydraulic fracturing. If casing integrity is compromised, acidizing or other less-intensive methods might be preferred.

Perforation Design: The size, density, and distribution of perforations in the wellbore casing determine the area of communication between the wellbore and the reservoir. Understanding the perforation design helps tailor the stimulation technique to effectively reach the target formation zone.

3. Production History:

Production Decline Curve: Analyzing the well's historical production data can reveal the nature of the decline. A sudden drop might indicate formation damage near the wellbore, while a gradual decline could suggest deeper issues requiring a more intensive stimulation approach.

Fluid Analysis: Studying the composition of produced fluids can provide clues about the type of formation damage. The presence of sand or scale can indicate the need for mechanical cleaning or acidizing, respectively.

4. Economic Considerations:

Cost-Effectiveness: Each stimulation method has associated costs, including mobilization, equipment, chemicals, and operational expenses. Evaluating the expected production gains needs to be balanced against the investment required for each technique.

Return on Investment (ROI): The chosen method should offer a positive ROI by significantly improving production rates at a cost that is outweighed by the additional revenue generated.

5. Additional Factors:

Environmental Regulations: Compliance with environmental regulations plays a role. Techniques with lower water usage or reduced surface footprint might be preferred in environmentally sensitive areas.

HSE Considerations: Health, Safety, and Environment (HSE) considerations are paramount. Techniques with lower risks of wellbore integrity issues or fluid leaks are often preferred

Challenges and Limitations

- **Inadequate Fluid Selection:** Incorrect fluid selection can worsen formation damage rather than mitigate it. For example, using incompatible chemicals can lead to precipitation or scaling in the formation.
- **Environmental Concerns:** Some chemicals used in stimulation washers, particularly acids and solvents, can have environmental impacts if not disposed of properly or if they leak into the surrounding environment.
- **Reservoir Heterogeneity:** Reservoirs with varying rock properties may require different stimulation techniques. The effectiveness of stimulation washers can vary depending on the geological complexity of the formation.
- **Overstimulation:** There is a risk that excessive washing or stimulation could damage the reservoir further, leading to the loss of valuable hydrocarbons.

POST-TREATMENT EVALUATION

After treatment:

- Recalculate PI
- Perform pressure buildup test
- Evaluate new skin factor
- Compare pre and post production

Successful treatment:

- Reduced skin
- Increased PI
- Higher stabilized production

INTEGRATED DAMAGE DIAGNOSIS WORKFLOW

- Identify production decline
- Compare with Arps model
- Calculate PI
- Conduct pressure test
- Diagnose damage mechanism
- Select wash or acidizing
- Execute treatment
- Evaluate performance

KEY ENGINEERING TAKEAWAYS

- Not all decline is reservoir depletion
- Skin factor is central to damage evaluation
- Near-wellbore damage has disproportionate effect
- Prevention is cheaper than remediation
- Proper diagnostics must precede stimulation

Why Formation Damage is Critical

Near-wellbore damage has a disproportionately large impact because: $\Delta P \propto \ln \left(\frac{r_e}{r_w} \right)$

Most pressure drop occurs close to the wellbore.

Therefore:

- 1–2 ft of damage can reduce production by 30–70%
- High skin factor dramatically reduces productivity index

Effect of Near-Wellbore Damage on Pressure Drop

An oil well produces from a reservoir with the following data:

- Permeability, $k=100$ md
- Thickness, $h=50$ ft
- Oil viscosity, $\mu=1.5$ cp
- Formation volume factor, $B=1.2$
- Drainage radius, $r_e=1000$ ft
- Wellbore radius, $r_w=0.5$ ft
- Production rate, $q=500$ STB/day

Question: Calculate the pressure drawdown (ΔP) required to produce 500 STB/day under steady-state radial flow conditions, assuming no formation damage ($S = 0$).

Solution:

Calculate the pressure drawdown (ΔP)

Step 1: Radial Flow Equation

For steady-state radial flow: $\Delta P = \frac{141.2 q \mu B}{kh} \ln\left(\frac{r_e}{r_w}\right) = \frac{141.2 \times 500 \times 1.5 \times 1.2}{100 \times 50} \ln\left(\frac{1000}{0.5}\right) = 25.416 \times 7.601 = 193.2 \text{ psi}$

Pressure Drop Without Damage: $\Delta P = 193.2 \text{ psi}$

Introduce 2 ft Damaged Zone

Suppose permeability near the wellbore (first 2 ft) drops from 100 md to 20 md.

This introduces skin effect.

Skin factor formula for damaged zone: $\left(\frac{k}{k_d} - 1\right) \ln\left(\frac{r_d}{r_w}\right)$

Where: $\left(\frac{100}{20} - 1\right) \ln\left(\frac{2}{0.5}\right) = 5.54$

• $k=100$

• $k_d=20$

• $r_d=2$

• $r_w=0.5$

Step 2: New Pressure Drop: $\Delta P =$ Modified radial equation: $\frac{141.2q\mu B}{kh} [\ln(\frac{r_e}{r_w}) + S] = 25.416 \times (7.601 + 5.54) = 334psi$

New Pressure Drop With Damage: $\Delta P=334$ psi

Comparison

Case	Pressure Drop
No damage	193 psi
With 2 ft damage	334 psi

Increase in Pressure Drop: $\frac{334 - 193}{193} \times 100 = 73\%$

Just 2 ft of damage increased pressure drop by 73%.

This proves:

Most pressure loss occurs near the wellbore due to logarithmic flow behavior.

Effect of Skin Factor on Productivity Index (PI)

Problem Statement

Reservoir pressure = 3000 psi

Flowing bottomhole pressure = 2500 psi

So:

$\Delta P=500\text{psi}$

Using previous reservoir properties:

$$PI = \frac{0.00708kh}{\mu B[ln(r^e/r_w) + S]}$$

Calculate PI Without Damage (S = 0)

$$PI = \frac{0.00708 \times 100 \times 50}{1.5 \times 1.2 \times 7.601} = \frac{35.4}{13.682} = 2.59 \text{ STB/day/psi}$$

Production Rate: $q=PI \times \Delta P = 2.59 \times 500=1295 \text{ STB/day}$

With Damage (S = 5.54)

$$1.8 \times (7.601+5.54)=1.8 \times 13.141=23.65$$

$$PI = \frac{35.4}{23.65} = 1.50 \text{ STB/day/psi}$$

New Production Rate

$$q=1.50 \times 500=750 \text{ STB/day}$$

Comparison

Case	PI	Production
No damage	2.59	1295 STB/day
With damage	1.50	750 STB/day

Production Reduction = $\frac{1295 - 750}{1295} \times 100 = 42\%$

FINAL ENGINEERING CONCLUSION

- 1. Because radial flow follows: $\Delta P \propto \ln \left(\frac{r_e}{r_w} \right)$
- 2. The logarithmic nature concentrates pressure drop near the wellbore.
- 3. A small damaged zone (1–2 ft) can:
 - Increase pressure drop by 70%+
 - Reduce production by 30–50%
 - Increase skin factor significantly

Formation damage is critical because the near-wellbore region controls the entire well productivity.